



**Service and Operational Optimization of AMRIT Pharmacy at
KGMU**

**BUSINESS DATA MANAGEMENT-CAPSTONE
PROJECT-FINAL REPORT**

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03rd Oct, 2025

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Declaration Statement

I am working on a Project titled “**Service and Operational Optimization of AMRIT Pharmacy at KGMU**”. I extend my appreciation to **Amrit Pharmacy**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

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Signature of Candidate

Name: Neeraj Kumar

Date: 03/10/2025

1. Executive Summary:

Amrit Pharmacy is located inside King George's Medical University (KGMU), Lucknow. It is known as a trusted and affordable place to buy medicines and medical supplies. Since the pharmacy buys medicines directly from manufacturers and runs within the hospital campus, it is able to cut down extra costs and provide medicines at cheaper rates. This also makes it easy for patients and visitors to get their medicines without going outside the hospital. Because of this, the pharmacy plays an important role in KGMU's healthcare system. But with the growing number of customers and thousands of sales every day, some problems like stock shortages, irregular sales, and slow billing have started creating difficulties in smooth functioning.

To study these problems in detail, a dataset of 56,281 rows and 21 attributes was collected, cleaned, and arranged properly. This data helped in analyzing sales according to medicines, categories, and time periods. The analysis showed that important medicines like Amlorin and Diamox had regular daily demand, but items like 3M Avagard Handrub had a lot of ups and downs because of bulk buying and returns. This shows that there is a need for better forecasting and planning of inventory so that important medicines are always available.

The category-wise study gave more information. Gastroenterology medicines showed steady growth in demand, orthopaedic items remained stable, while cardiology medicines showed a small decline. Time-based analysis showed that sales were highest on Mondays and Tuesdays, while Sundays had the lowest sales. These changes in demand proved that stock levels and staff management should be adjusted according to weekly sales patterns. ABC segmentation was also done to divide products into high, medium, and low priority groups. This will help in keeping a balance between stock optimization, cost control, and expiry management.

In the end, the analysis makes it clear that Amrit Pharmacy has done well in keeping medicines affordable and easily available, but some operational improvements are required for better efficiency. If the pharmacy adopts proper forecasting methods, improves billing, and manages stock more effectively, it can avoid disruptions and give even better services. With these changes, Amrit Pharmacy can continue to support KGMU by ensuring patients and visitors always get reliable, affordable, and timely access to medicines.

2. Detailed Explanation of Analysis Process/Method:

2.1. Data Acquisition:

My connection with the pharmacy's employee, Mr. Arvind, actually came from a personal experience. Around two years ago, when I was unwell and receiving treatment at King George's Medical University (KGMU), I visited the pharmacy and met him there. That short interaction stayed in my mind. So, when I started working on this project, I decided to contact him again, as he was directly linked to the pharmacy and could help me get the data I needed.

When I first contacted Mr. Arvind, he was understandably hesitant to share the pharmacy's data with me, as it involved sensitive and internal information. However, I explained the academic purpose of my project in detail, making it clear that the data would be used purely for research and analysis. I also mentioned that I am an IIT, madras student and sent him a copy of my BDM bonafide certificate as proof of authenticity. This reassured him of the legitimacy of my request, and eventually, he agreed to provide me with the data required for my study.

In order to formally document this process and add credibility to my project, I conducted a Google Meet session with Mr. Arvind. This virtual meeting served as proof of our interaction and became an important part of validating the source of my data collection.

The data collection itself took place over a span of three months, from March 1 to May 31, 2025. The pharmacy provided me with their system-generated data in its raw form, stored in an Excel file. The dataset was quite large, containing a total of 56,281 individual bills during this period.

My next step involved processing this raw data into a form that could be meaningfully analyzed. This required digitization and consolidation. I systematically aggregated the daily sales of each item, so that instead of having thousands of separate bills, each day's sales for a particular item were recorded in a single row.

2.1.1 Field Names:

SL.NO	Branch Code	Branch Name	Date	Invoice No	ITEM NAME	Pack	Item Category	Item Group
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BATCH NO	Quantity Per Box	SALE PK QTY	SALE LS QTY	MRP	SELLING RATE (WITHOUT TAX)	E.P.R	E.P.R VALUE	Basic Selling Price
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Item Gst %	Tax Amt	HSN Code	Sale Value	Mrp Value	Discount	Discount %	Cust Code	Cust Name	Payment Type
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2.2. Data Cleaning and and Preprocessing:

The sales data for Amrit Pharmacy was initially received in a raw, system-generated format, which required significant cleaning and consolidation before it could be analyzed. I started with a dataset containing 28 columns and 17,833 in rows (in March), 28 columns and 17,912 in rows (in April) and 28 columns and 20,536 in rows (in May) ,this is data of spanning of three-month period from March 1 to May 31, 2025.

During the initial review, I identified several issues. The raw data had missing fields, duplicate rows, and unwanted columns that were not relevant to the analysis. To address these problems and ensure accuracy, I performed a thorough data cleaning process.

In the cleaning process I used pandas and matplotlib libraries of python. Firstly remove the null and duplicate values from the data.

After this I aggregated daily sales for each item into a single summary row, merging multiple entries of the same product on the same day. This transformation from raw data to a structured format was crucial as it made it easier to recognize sales trends, spot fast-moving or frequently returned products, and track demand fluctuations. This organized dataset

enabled me to make more informed decisions regarding inventory management, procurement, and billing, which ultimately leads to better operational efficiency and service delivery.

After that I categorize the fields into the numerical and categorical categories so that overall analysis is easy. After that I used a describe function using pandas to get the count, mean, min, 25th, 50th, 75th percentile, max, std of the numerical category data.

After that I make a field using this formula

$$\text{Total quantity sold} = \text{Sale pk qty} * \text{Quantity per box} + \text{Sale ls qty}$$

To get the top items from different groups for this I categorize the data into multiple categories and then use the sum function from the pandas.

2.3. Data Analysis:

Data analysis for Amrit Pharmacy involved a multi-faceted approach to gain insights into the business's performance and operational efficiency. The methods used included descriptive statistics, trend analysis, and ABC segmentation to provide a comprehensive view of the sales and inventory data.

2.3.1. Descriptive Statistics Analysis:

When I worked with the **numeric data**, I looked at things like the average (mean), middle value (median), variation (standard deviation), skewness, and percentiles. Doing this gave me a good sense of how sales usually behaved, how much they went up and down, and whether certain products had steady demand or unusual spikes. I used the **describe()** function in Pandas for most of this, which made it easier to get these numbers quickly.

For the **categorical data**, I checked how many times different values appeared, how many unique options there were, and which ones showed up the most often. This helped me see which medicines were sold the most, which categories were common, and even which pack sizes customers preferred. Again, the **describe()** function was really helpful here because it summarized all this information in a simple way.

By breaking the analysis into numeric and categorical parts, I was able to see the sales trends more clearly. The numeric side showed me how sales moved over time, while the

categorical side pointed out which specific medicines and categories were driving those numbers. Together, this helped me figure out which medicines needed close stock monitoring and which ones sold less often and could be managed more carefully.

2.3.2. Trend Analysis:

To analyze the sales trends, I first arranged the dataset in a way that connected each transaction with its respective date. This allowed me to group the data by days, weeks, and months, which made it easier to notice patterns at different time levels. After preparing the data, I used Python's Matplotlib library to make line and bar charts. These visualizations helped me clearly see how sales were changing over time, rather than just going through raw figures.

I looked at the sales from multiple angles. The daily grouping showed the trend of small ups and downs, the weekly view gave a smoother idea of customer activity, and the monthly grouping revealed the broader trend. By comparing these different time frames, I was able to study the sales in a structured way. The combination of organizing the data and using different graphs made the overall analysis much easier to understand for me.

2.3.3. ABC Segmentation Analysis:

To study how different products were performing, I used the ABC analysis method, which groups items based on how much they contribute to total revenue of the pharmacy. I started by calculating the sales revenue for each medicine and then sorted them from highest to lowest revenue. After that, I calculated their cumulative percentage contribution to identify which products were bringing in the most revenue of the pharmacy.

Using the standard ABC approach, I divided the medicines into three category:

Category A: High-priority medicines contributing around 70-80% of total revenue.

Category B: Medium-priority items contributing around 20-30%.

Category C: Low-priority medicines contributing only about 5-10%, usually slow-moving or less demanding medicines.

This grouping helped me to understand which medicines needed to be restocked frequently, which ones required moderate attention, and which could be kept in smaller quantities. Overall, the ABC analysis proved to be a simple and practical method for

managing inventory, preventing overstocking, and making sure that high-demand medicines were always available in the inventory of the pharmacy.

2.4. Analysis Techniques Used:

The data of Amrit Pharmacy was examined using simple mathematical calculations, basic statistical methods, and data validation techniques. These approaches helped me to understand the useful insights from the data and supported better decision-making to improve the pharmacy's overall operations and performance.

2.4.1. Sales Analysis Techniques:

Descriptive Statistics: To study sales patterns and changes, I used descriptive statistics such as mean, median, minimum, maximum, standard deviation, kurtosis, and skewness. These calculations helped me understand the average sales, spot fluctuations, and know me to unusual trends. By looking at these numbers, I could see sales trends over the time, including peak days and monthly variations in the pharmacy's sales.

Trend Analysis: I also carried out trend analysis to examine sales on a daily, weekly, and monthly basis. Using line and bar charts with the help of Python's matplotlib library, I was able to visualize performance over time, identify recurring patterns, and spot unusual sudden changes. This made it easier to see when demand was high or low, which is helpful for managing staff, planning promotions, and organizing resources effectively so that we can manage things efficiently.

2.4.2. Inventory Analysis Techniques:

ABC Segmentation: To manage inventory more efficiently, I used the ABC analysis method, which ranks products according to how much they contribute to total revenue of the pharmacy.

The medicines were divided into three categories:

Category A: High-priority, high-revenue items

Category B: Medium-priority products

Category C: Low-priority or slow-moving items

Descriptive Statistics: I also used descriptive statistics to study the sales patterns of individual medicines. This helped identify which medicines had consistent daily demand and which showed irregular sales. Using this information, inventory could be managed more effectively: high-demand medicines were kept well-stocked, while low-demand items were maintained in smaller quantities to prevent overstocking.

3. Results and Findings:

The analysis of Amrit Pharmacy's data generated valuable insights into operational efficiency, product demand, and inventory management practices. By applying descriptive statistics, trend analysis, and ABC segmentation, sales patterns were examined across daily, weekly, and monthly intervals. This made it possible to identify high-demand and low-demand products and to assess the revenue contribution of each item. These insights form a strong foundation for recommendations aimed at enhancing inventory planning, optimizing stock levels, and improving the overall effectiveness of the pharmacy's operations.

3.1. Sales Pattern Analysis:

To gain a deeper understanding of Amrit Pharmacy's sales performance, the data was analyzed across three timeframes: daily, weekly, and monthly. This multi-level approach provided clarity on both short-term fluctuations and long-term demand trends. It helped in identifying recurring patterns, unusual variations, and peak as well as low-activity periods. Considering all three perspectives together allowed the analysis to capture immediate shifts while also highlighting broader trends, offering valuable insights for decisions related to inventory management, workforce planning, and promotional strategies.

3.1.1. Daily Sales Trend:

The daily sales analysis highlighted the variation in sales from one day to the next. The line chart showed noticeable fluctuations, with certain days experiencing high demand while others recorded sharp declines. These changes may be influenced by factors such as stock availability, customer purchasing behavior, or regular buying cycles. Monitoring daily sales is essential, as it helps identify unusual trends and enables more effective planning to ensure a consistent and reliable inventory flow.

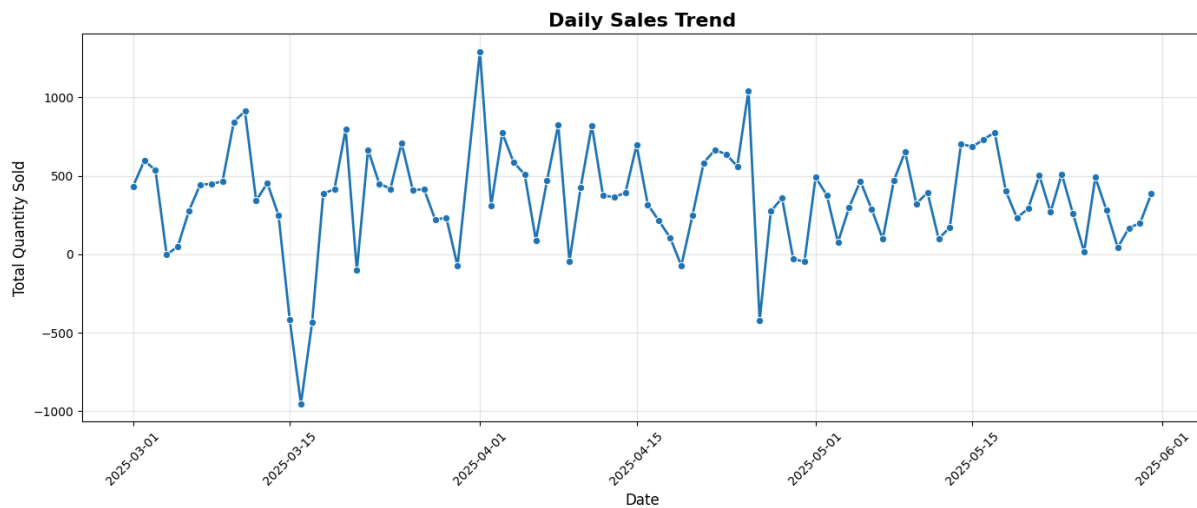


Figure1 : Daily Sales Trend

3.1.2. Weekly Sales Trend

The weekly analysis showed regular patterns in sales throughout the week. Mondays, Tuesdays, and Fridays had higher sales, while Sundays usually had the lowest. This indicates that weekdays are the busiest, so the pharmacy needs enough staff and stock to meet demand. Slower days like Sundays could be used for special offers or promotions to attract more customers.

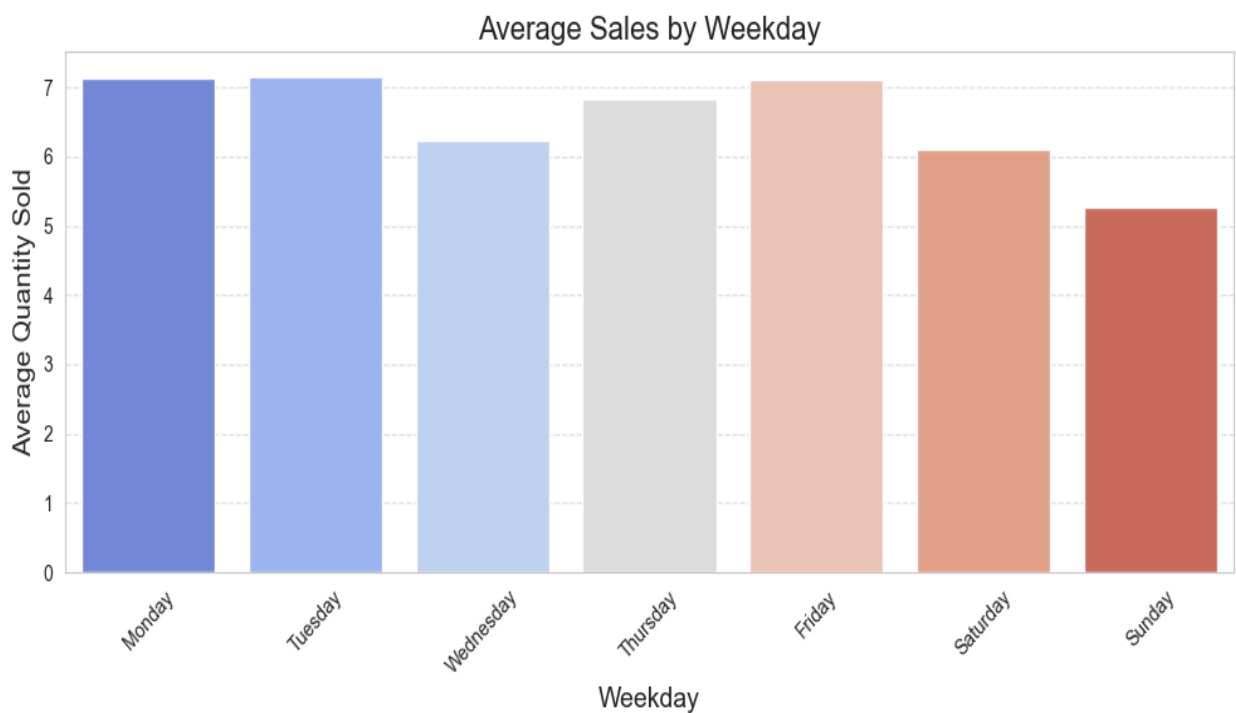


Figure 2 : Weekly Sales Trend

3.1.3. Monthly Sales Trend

Looking at sales on a monthly level, March, April, and May showed steady performance, with March recording slightly higher totals than April and May. Overall, the differences between the three months were minimal, indicating that customer demand was fairly consistent during this period without any major seasonal shifts.

Sales showed a slight downward trend after March, with April and May experiencing small declines. However, the overall sales volumes remained strong and stable, reflecting predictable demand. The consistent performance highlights a balanced market, where customer purchases stayed reliable without major fluctuations.

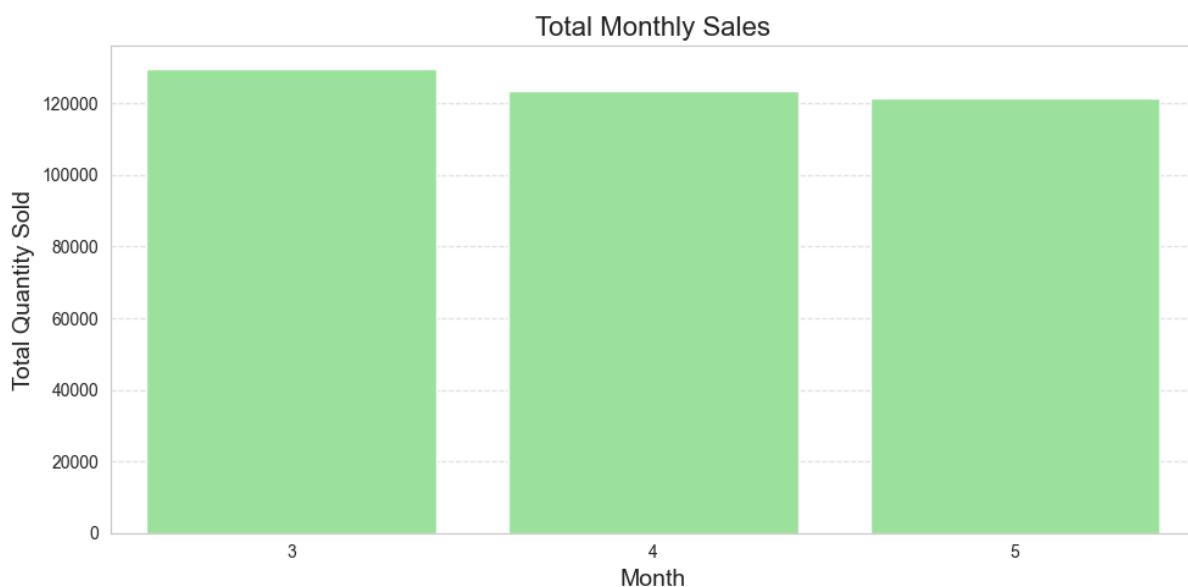


Figure 3 : Monthly Sales Trend

3.2. Product Performance Analysis:

To evaluate the overall performance of products at Amrit Pharmacy, the analysis was carried out in three main areas: identifying the best-selling medicines, highlighting slow-moving or low-demand items, and examining sales distribution across pack sizes and product categories. This multi-dimensional approach helps in understanding which products are contributing the most to sales, which are creating challenges in inventory turnover, and how customer preferences differ by product type and packaging.

3.2.1. Top-Selling Medicines

The analysis of top-performing products revealed a group of medicines and medical supplies that consistently drive sales. Among them, BUDERATOR RESPULES [IB4787] and NOVOM INJ [IW9798] emerged as clear leaders, each recording more than 1,200 units sold. Close behind is TRAMAKACH INJ [IU9356], also crossing the 1,200-unit mark, reflecting strong and regular demand. Beyond these, essential supplies such as the 20 ML SYRINGE (NEEDLE -21G X 1 1/2") [ID1413], with over 800 units sold, underscore the steady need for basic healthcare items. These results highlight the importance of keeping high-demand products adequately stocked, as they not only generate revenue but also play a crucial role in uninterrupted patient care.

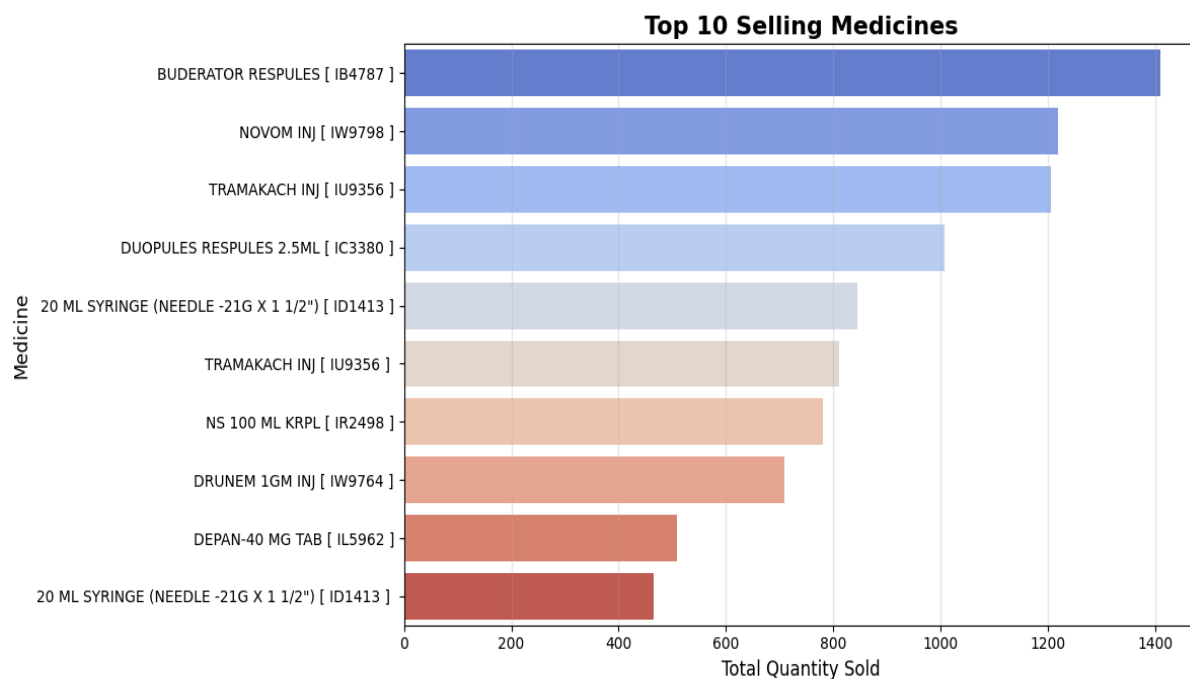


Figure 4 : Top 10 Selling Medicines

3.2.2. Low-Demand or Slow-Moving Products

In contrast, the analysis also highlighted products with low or even negative sales, indicating challenges such as weak demand, product returns, or inventory adjustments. The slowest-moving item was STENIVITA -GOLD TAB [IN7428], which reported negative sales. Similarly, items like EXAMINATION DELSWEEFT GLOVES 100 [IP8808] and RB-CAL TABLET [IC5481] showed very limited movement, signaling

potential inefficiencies in stock management.

These patterns point to possible overstocking, restricted patient demand, or the need for stronger promotional efforts to improve turnover. Identifying such slow movers is critical for optimizing procurement strategies, lowering storage expenses, and preventing unnecessary capital from being tied up in underperforming products.

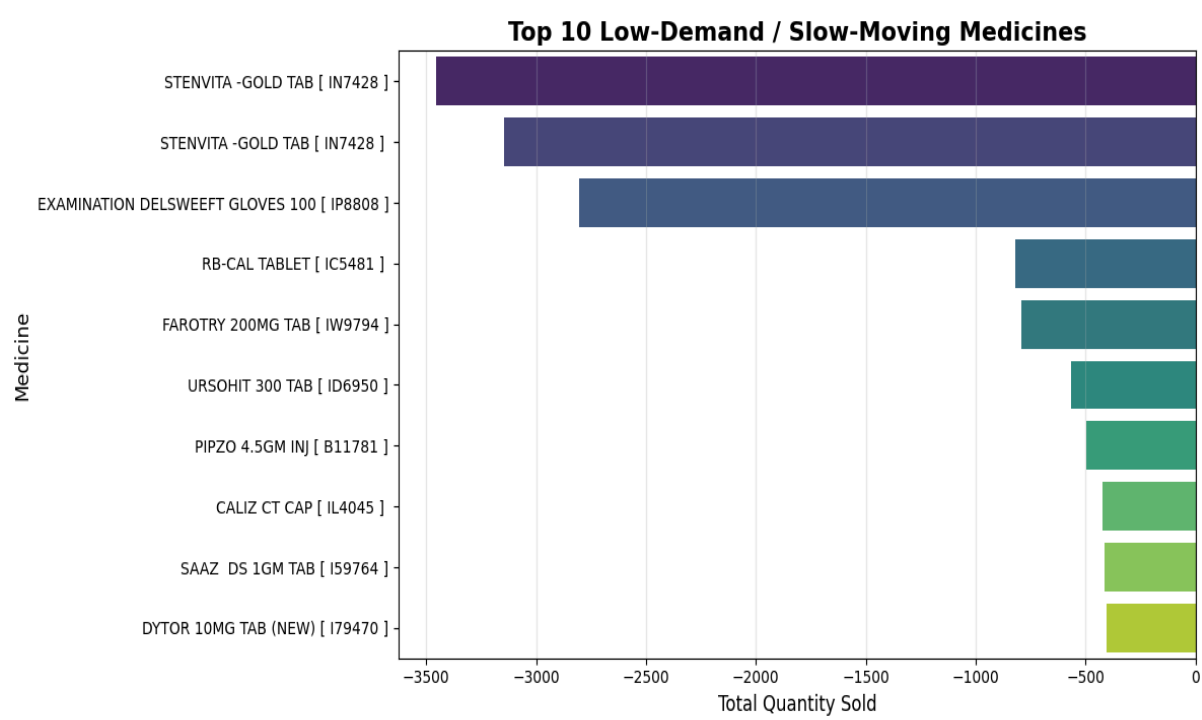


Figure 5 : Top 10 Slow Moving Medicines

3.2.3. Pack Size and Category Preferences

An additional layer of analysis examined sales by pack size and product category, providing insights into customer purchasing behavior. The results showed a clear preference for smaller pack sizes, particularly 1X1 and 2ML, which significantly outperformed larger options, reflecting a tendency for patients and healthcare providers to choose more practical and affordable quantities aligned with immediate treatment needs. At the category level, General Medicine and General Surgery accounted for the majority of sales, establishing themselves as the backbone of pharmacy operations, while categories such as Rheumatology and Endocrinology recorded relatively low volumes, suggesting the need for careful stocking in smaller quantities to balance availability with reduced risk of overstocking and wastage.

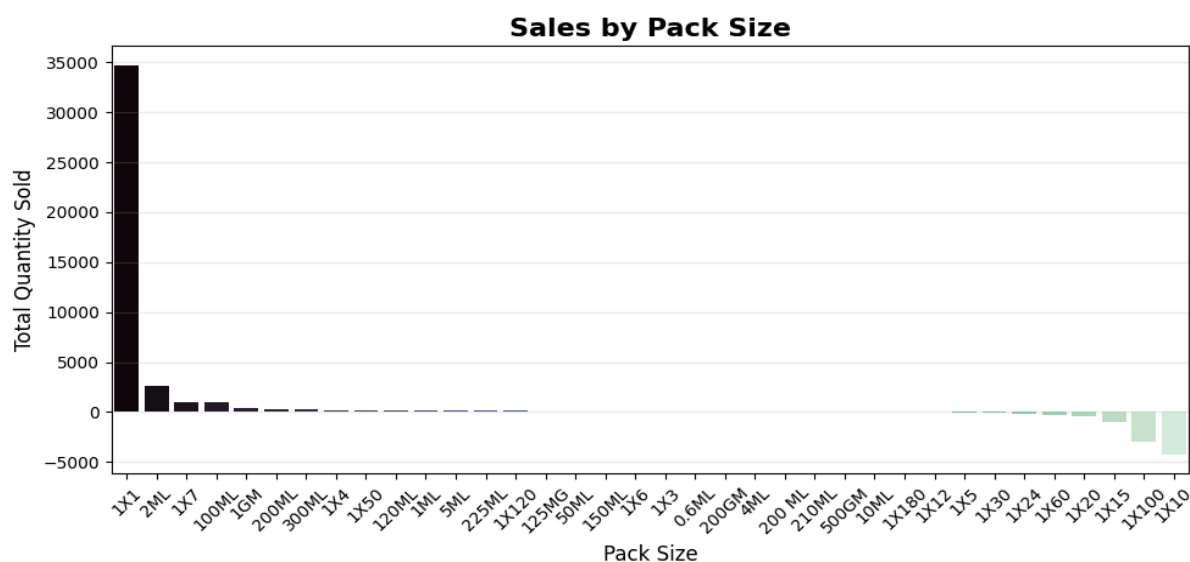


Figure 6 : Sale by Pack Size

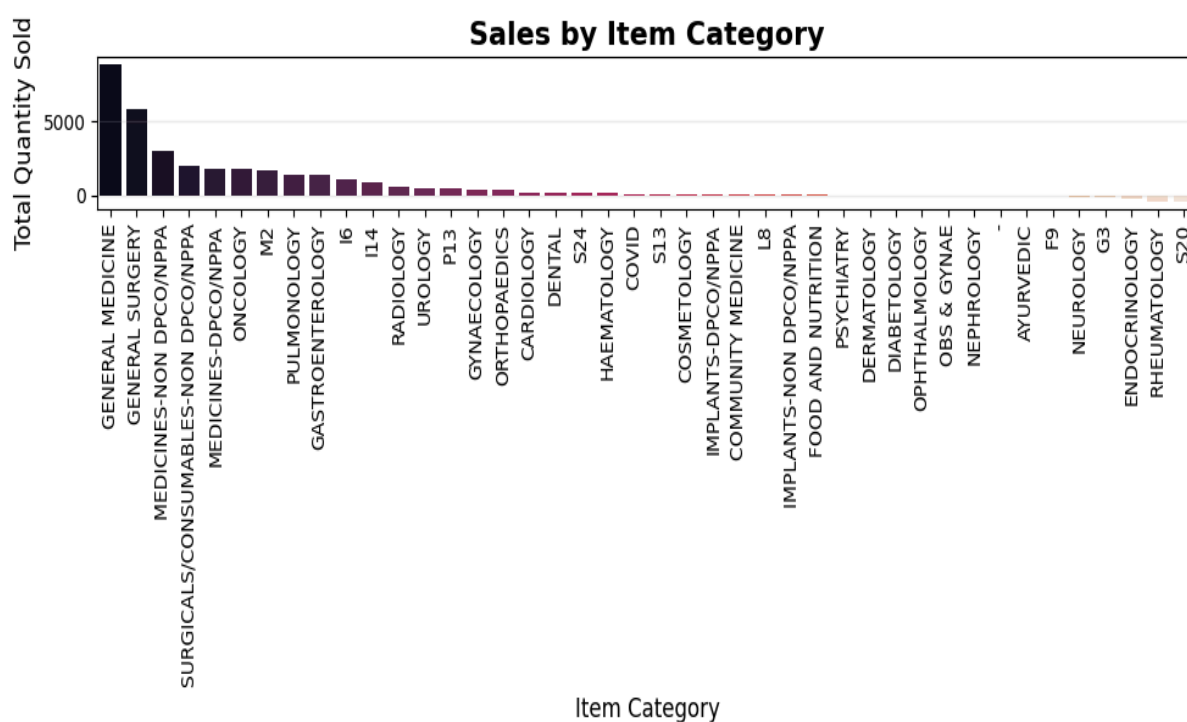


Figure 7 : Sale by Item Category

3.3. Inventory Segmentation

An inventory segmentation analysis was carried out to optimize stock management and improve the efficiency of resource allocation. By applying the ABC analysis method, the entire product range of Amrit Pharmacy was divided into three categories based on each item's contribution to total revenue. This classification provided a practical framework for identifying which items required the highest level of attention and which could be managed with a more flexible approach.

3.3.1. ABC Analysis: High, Medium, and Low-Priority Items

The analysis showed that Category A contained 237 items, which accounted for nearly 70–80% of the total revenue despite representing a small portion of the overall inventory. These are high-priority products that must always be available, as they have the greatest financial impact. Category B included 350 items, contributing the next 20–25% of revenue, making them moderately important. The largest group, Category C, consisted of 790 items, which collectively generated less than 10% of revenue. Although they form the bulk of the inventory in terms of count, their individual sales impact was minimal.

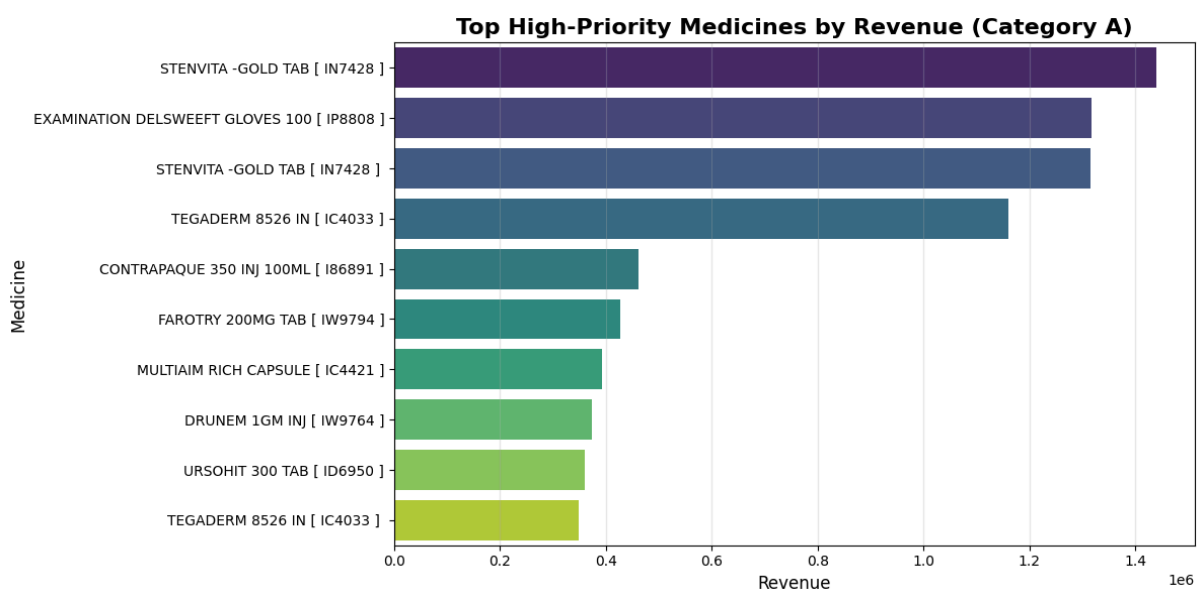


Figure 8 : Top High Revenue Medicines

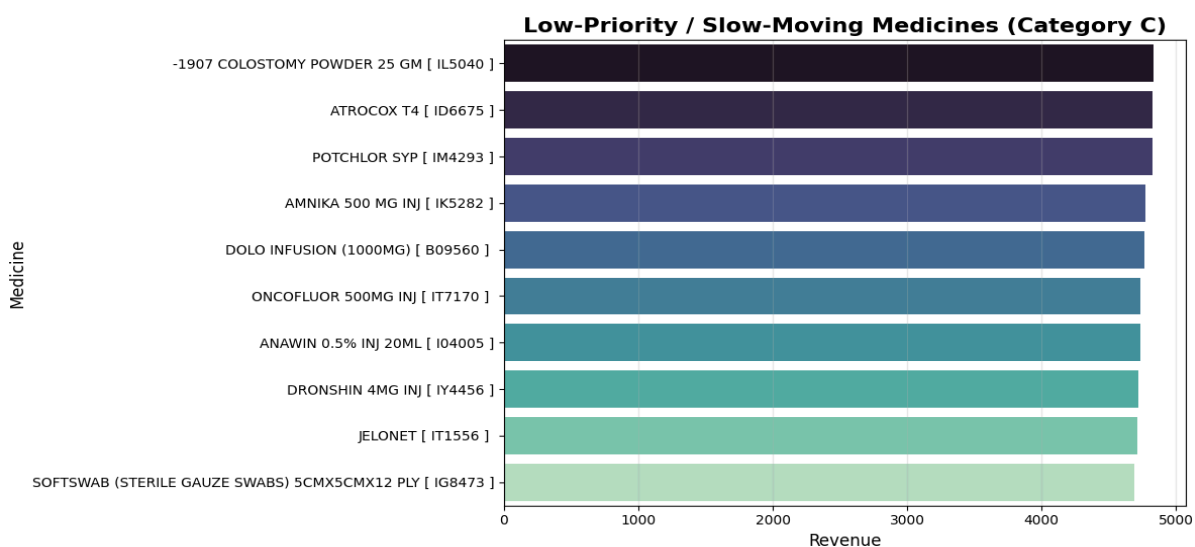


Figure 9 : Top Low Revenue Medicines

3.3.2. Stock Replenishment Priorities

The results indicate that Category A products require strict monitoring and frequent replenishment due to their high demand and revenue contribution. Any shortage in this segment would directly affect both sales and customer satisfaction. Products in Category B require a balanced approach, ensuring steady availability while avoiding unnecessary overstocking. For Category C, the strategy should focus on maintaining only minimal stock levels, since over-purchasing these items could lead to wastage and expiry.

3.3.3. Identifying Overstocking Risks

When looking more closely at Category C items, it became clear that many of them carry a higher risk of overstocking and don't move as quickly as other products. These tend to be specialized medicines or low-demand supplies that contribute little to overall revenue but still take up shelf space. A practical way to manage this is by placing smaller, less frequent orders and using promotions or clearance sales to reduce the existing stock. This not only helps prevent losses from expired items but also frees up cash that would otherwise be tied up in products that don't sell well.

3.4. Revenue and Profitability Insights

The financial review of Amrit Pharmacy shows where most of the revenue comes from, how discounts affect sales, and which products are actually the most profitable. The findings suggest that while only a few categories and products bring in the bulk of sales, the higher profits often come from niche or high-margin items that aren't always the top sellers. These insights give a solid foundation for making smarter decisions around pricing, discount strategies, and how inventory should be managed.

3.4.1. Revenue Contribution by Product and Category

The revenue distribution across categories shows a significant concentration in a few key areas. General Medicine is the leading category with more than ₹10 million in revenue, accounting for the bulk of overall sales. This is followed by Surgicals/Consumables at approximately ₹2.5 million and Medicines – Non DPCO/NPPA contributing over ₹2 million. These figures confirm that general healthcare and surgical supplies form the financial backbone of the pharmacy.

At the individual product level, the contribution is similarly skewed. STENIVITA-GOLD TAB [IN7428] generated about ₹1.48 million, while EXAMINATION DELSWEEFT GLOVES 100 [IP8808] added ₹1.46 million. Other high-value items include TEGADERM 8526 IN [IC4033] at ₹1.1 million and CONTRAPAQUE 350 INJ 100ML [I86891] at ₹0.42 million. Together, these products form a small fraction of the inventory but deliver a disproportionately large share of revenue, underlining the importance of consistently maintaining their stock levels.

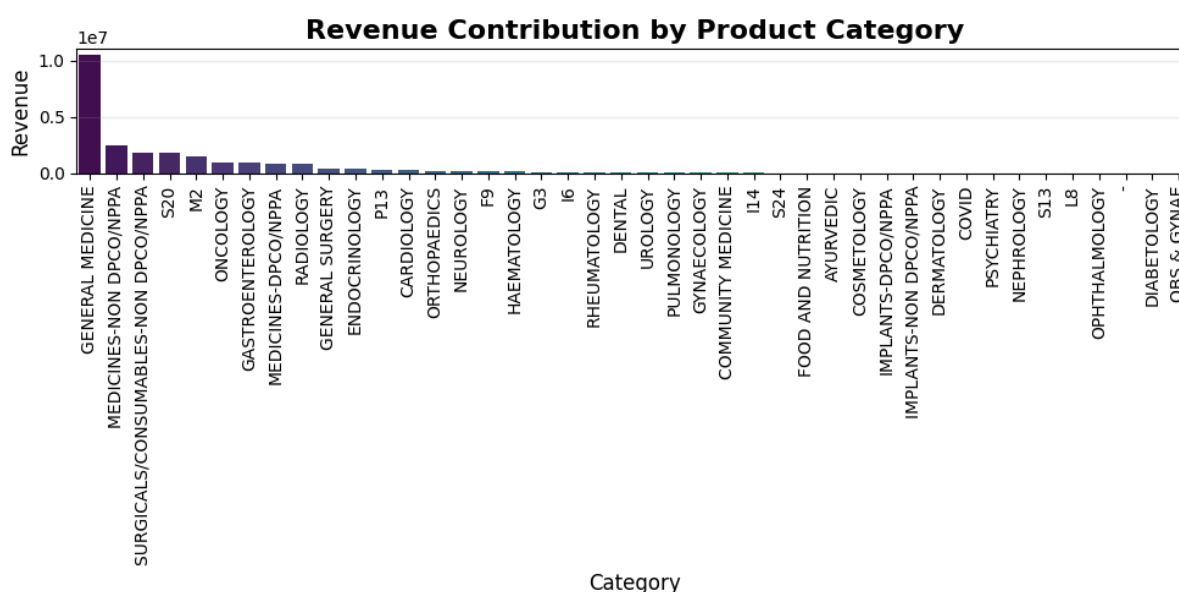


Figure 10 : Revenue Contribution by Product Category

3.4.2. Discount and Price Impact on Sales

The analysis of discounts revealed that a majority of revenue was earned at zero or very low discount rates, showing strong baseline demand for many essential products. However, some transactions applied discounts ranging from 2% up to over 150%, particularly in high-revenue categories such as General Medicine and Surgicals/Consumables. The wide variation in discount levels suggests that while discounts can stimulate demand, their impact is inconsistent. In some cases, aggressive discounting on high-value products inflated revenue figures, but such practices also raise concerns about eroding margins. This indicates the need for a structured discounting strategy that balances sales growth with profitability protection.

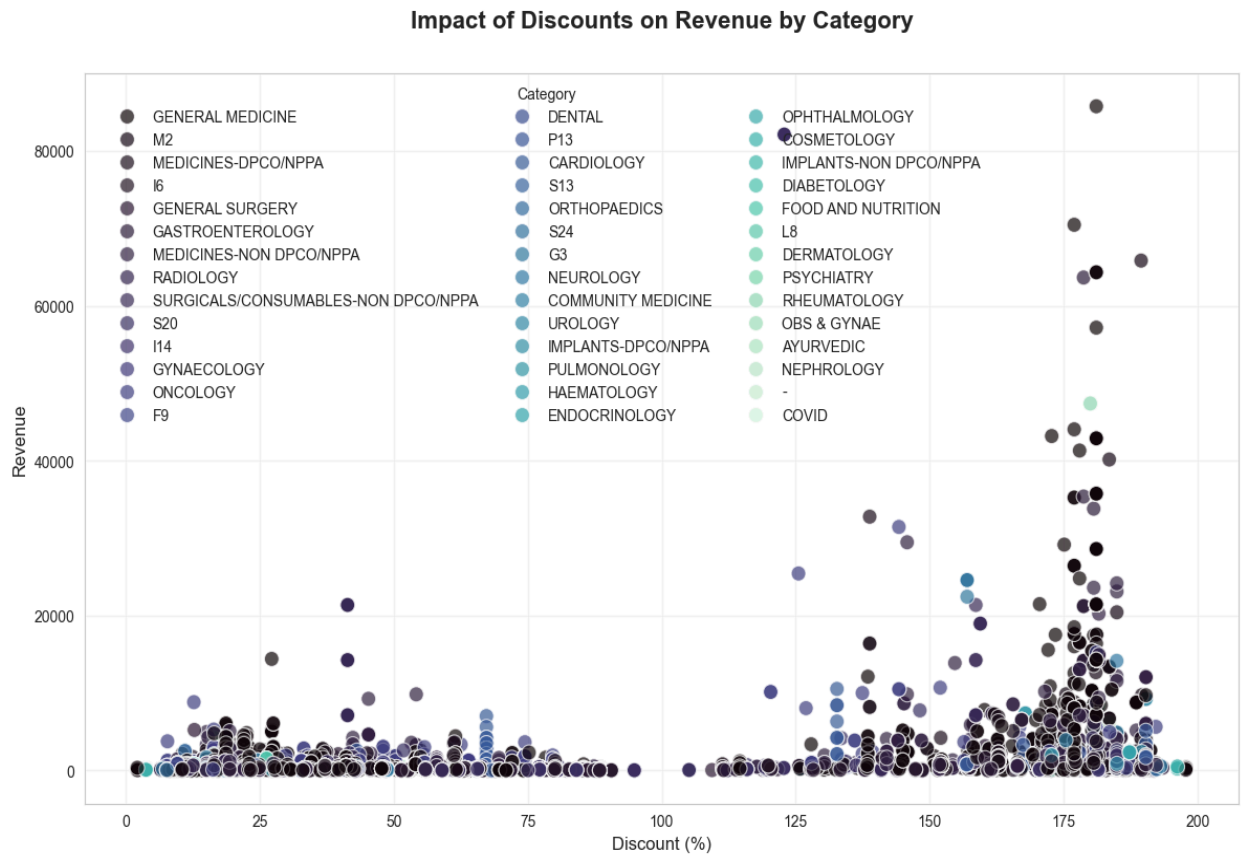


Figure 11 : Impact of Discounts on Revenue by Category

3.4.3. Profit Margins and High-Revenue Products

Looking at profit margins shows a clear difference between products that drive revenue and those that actually deliver profitability. High-selling items like STENIVITA-GOLD TAB and EXAMINATION DELSWEEFT GLOVES are important for keeping sales volume steady, but their margins are relatively low. On the other hand, products such as TEGADERM 8526 IN [IC4033] and various TRANSPORE SURGICAL TAPE variants consistently generated margins close to 100%, making them highly profitable even with smaller sales volumes. This contrast highlights the need for balance—using high-revenue products to support steady turnover while also focusing on high-margin items that strengthen overall profitability.

4. Interpretation of Results

4.1. Sales and Demand Patterns

Analysis of sales across daily, weekly, and monthly levels revealed consistent monthly

performance but clear weekday peaks. Mondays, Tuesdays, and Fridays showed higher activity compared to Sundays, which recorded the lowest sales. These patterns indicate predictable demand cycles that can guide staffing and promotional planning.

4.2. Product Performance Insights

Product-level analysis highlighted both high-demand and low-demand items. Medicines such as BUDERATOR RESPULES [IB4787], NOVOM INJ [IW9798], and TRAMAKACH INJ [IU9356] each sold more than 1,200 units, marking them as consistent top performers. On the other hand, items like STENIVITA -GOLD TAB [IN7428] and EXAMINATION DELSWEEFT GLOVES 100 [IP8808] showed low or even negative sales, indicating overstocking risks or limited customer demand.

4.3. Inventory Segmentation and Stock Priorities

ABC segmentation of the inventory revealed that Category A (237 items) accounted for nearly 70-80% of total revenue, making them high-priority for replenishment and stock monitoring. Category B (350 items) contributed moderately, while Category C (790 items) had minimal revenue impact but carried higher risk of overstocking and expiry. This segmentation provides a framework for prioritizing inventory investments.

4.4. Revenue and Profitability Insights

Revenue contribution analysis showed that General Medicine generated over ₹10 million, making it the strongest category, followed by Surgicals/Consumables (₹2.5 million) and Non-DPCO/NPPA Medicines (₹2 million). At the product level, STENIVITA -GOLD TAB [IN7428] and EXAMINATION DELSWEEFT GLOVES 100 [IP8808] each contributed more than ₹1.4 million in revenue. Profitability analysis further revealed that surgical consumables like TEGADERM 8526 IN [IC4033] had margins close to 100%, underlining their importance despite lower sales volume.

5. Recommendations for the Problem

Based on the analysis of the data and the data trends, the following recommendations are proposed to address the key challenges identified at Amrit Pharmacy

5.1. Optimize Inventory for High-Demand and Profitable Categories

The analysis shows that certain categories consistently contribute higher sales and profitability. Products with strong demand and higher profit margins should be prioritized in inventory planning to avoid frequent stock-outs. Regular tracking of fast-moving items will help maintain availability and ensure steady revenue growth.

5.2. Reassess Discount Strategies

The impact of discounts on sales reveals that while promotions can temporarily increase sales volume, excessive discounting reduces overall profitability. It is recommended to adopt a data-driven discounting policy by focusing discounts only on slow-moving products or seasonal items. Controlled and targeted discounts will ensure revenue growth without compromising margins.

5.3. Strengthen Digital Payment Adoption

Sales analysis by payment type indicates a significant portion of transactions are shifting towards digital methods. Encouraging digital payments can improve billing efficiency, reduce cash-handling risks, and provide cleaner transaction records for financial tracking. Offering small incentives for digital payments may further increase adoption.

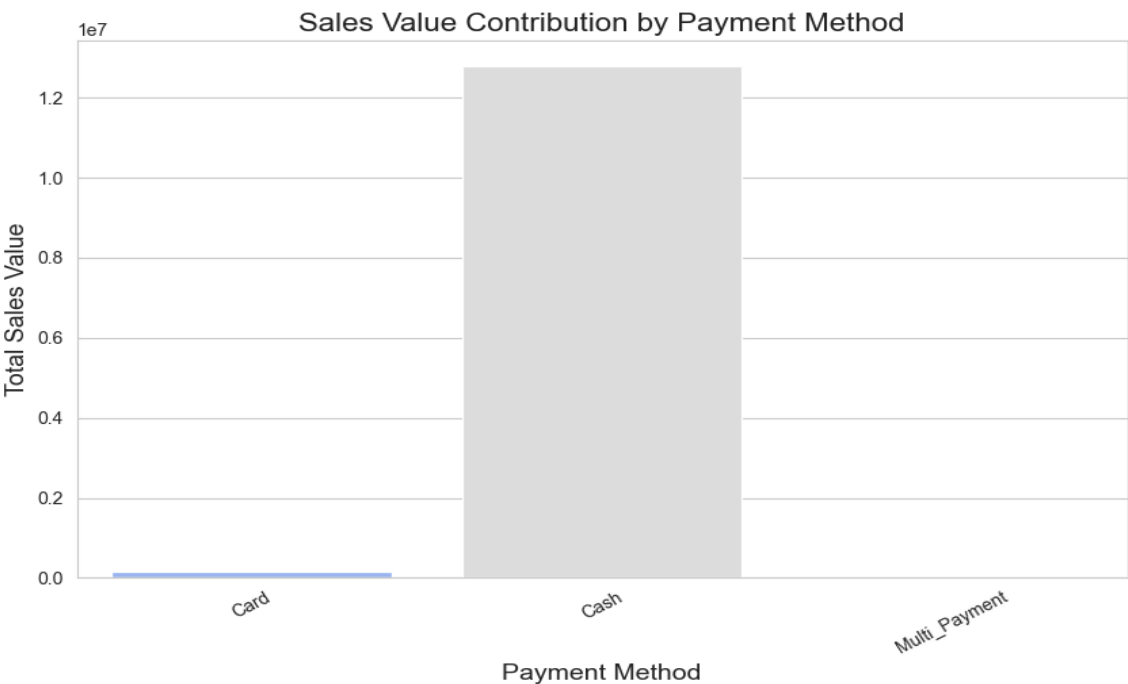


Figure 12: Sales value Contribution by Payment method

5.4. Implement Data-Driven Forecasting and Reporting

A structured sales trend and profitability analysis should be integrated into regular operations. Automating sales reports, category-wise profitability dashboards, and monthly performance reviews will help management make quicker, evidence-based decisions.

5.5. Strategies to Minimize Waste for High-Stock, Low-Sales Items

The analysis of pharmacy inventory data has revealed a distinct group of items with high stock levels but low sales performance, as illustrated in the accompanying chart. Notably, products such as "EXAMINATION DELSWEEFT GLOVES 100" represent a disproportionately high portion of total inventory value, while their turnover remains low. This imbalance signals an urgent need for intervention to avoid excessive waste and optimize inventory management practices. By focusing efforts on these specific items, the pharmacy can realign its stock with demand patterns, reduce carrying costs, and ultimately ensure that resources are directed toward products with higher sales potential and lower risk of expiration or obsolescence.

Dataset Link:  final_dataset.xlsx

Analysis Code File link:  code_file